

Lab 6.1 — Wokwi ESP32 + DHT22 + HiveMQ / MQTTX

Aluno: Thales T | Data: 21/09/2026
Broker: broker.hivemq.com | Wokwi TCP 1883 | MQTTX wss :8884 /mqtt
Topicos: aulas/professortupi/temperatura e aulas/professortupi/umidade
Circuito: DHT22 VCC->3V3, GND->GND, SDA->GPIO14 | WiFi Wokwi-GUEST
STATUS: PASS — MQTTX recebeu temperatura 80.00 e umidade 72.00

Print MQTTX (publicacoes no broker):

Connections

testemqtt@broker.hivemq.co...

testemqtt

Name

testemqtt

Client ID

mqttx_250ba801

Username

Password

Keep Alive

60

Clean Session

☒ true

Disconnect

+ New Subscription

aulas/professortu... QoS 0

Plaintext

72.00
2026-09-21 16:34:12:096
Topic: aulas/professortupi/temperatura QoS: 0
80.00
2026-09-21 16:34:17:734
Topic: aulas/professortupi/umidade QoS: 0
72.00
2026-09-21 16:34:17:949

Print Wokwi (circuito ESP32 + DHT22):

WOKWI

SAVE

SHARE

Circuito DHT22: VCC--3V3, GND--GND, SDA--GPIO 14

Docs

sketch.ino

diagram.json

Library Manager

libraries.txt

```
1 #include <WiFi.h>
2 #include <PubSubClient.h>
3 #include <DHT.h>
4
5 // --- Hardware e Wi-Fi (Wokwi) ---
6 #define DHT_PIN 14
7 #define DHT_TYPE DHT22
8 #define WIFI_SSID "wokwi-guest"
9 #define WIFI_PASS ""
10
11 // --- MQTT (broker público HiveMQ) ---
12 #define MQTT_SERVER "broker.hivemq.com"
13 #define MQTT_PORT 1883
14 #define CLIENT_ID "Projeto_IOT_WiFi_Prof_Tupi_087"
15
16 #define TOPIC_TEMP "aulas/professortupi/temperatura"
17 #define TOPIC_HUM "aulas/professortupi/umidade"
18
19 DHT dht(DHT_PIN, DHT_TYPE);
20 WiFiClient espClient;
21 PubSubClient mqttClient(espClient);
22
23 unsigned long lastMsg = 0;
24 const long PUBLISH_INTERVAL = 5000;
25
26 void setup_wifi() {
27   delay(10);
28   Serial.print("Conectando a rede: ");
29   Serial.println(WIFI_SSID);
30
31   WiFi.begin(WIFI_SSID, WIFI_PASS);
32
33   while (WiFi.status() != WL_CONNECTED) {
34     delay(500);
35     Serial.print(".");
36   }
37
38   Serial.println("\nWiFi Conectado!");
39   Serial.print("Endereço IP: ");
40   Serial.println(WiFi.localIP());
41 }
42
43 void reconnect_mqtt() {
44   while (!mqttClient.connected()) {
45     Serial.print("Tentando reconectar MQTT: ");
46     Serial.print(mqttClient.connect("Projeto_IOT_WiFi_Prof_Tupi_087", "", ""));
47     Serial.println();
48     delay(1000);
49   }
50 }
```

Simulation

02:27.216 73%

Editing DHT22

12 KB 6.8 KB Public

Temperature:

Humidity:

ENABLE PRIVATE GATEWAY

Download PCAP file

WiFi Documentation

ESP32

DHT22